Case III. When roots of indicial equation are distinct and differ by an integer, making a coefficient of y infinite.

Let m_1 and m_2 be the roots such that $m_1 < m_2$. If some of the coefficients of y series become infinite when $m = m_1$, we modify the form of y by replacing a_0 by $b_0\,(m - m_1)$. Then the complete solution is

$$y = C_1 (y)_{m_2} + C_2 \left(\frac{\partial y}{\partial m}\right)_{m_1}$$

Example *Obtain the series solution of the equation*

$$x(1 - x)\frac{d^2y}{dx^2} - (1 + 3x)\frac{dy}{dx} - y = 0.$$

Solution. Here $x = 0$ is a singular point, since coefficient of y'' is zero at $x = 0$.

$\therefore$ substituting $\quad y = a_0x^m + a_1x^{m+1} + a_2x^{m+2} + \ldots$

$$\frac{dy}{dx} = ma_0x^{m-1} + (m + 1)a_1x^m + (m + 2)\,a_2\,x^{m+1} + \ldots$$

$$\frac{d^2y}{dx^2} = m(m - 1)\,a_0x^{m-2} + (m + 1)ma_1\,x^{m-1} + (m + 2)\,(m + 1)\,a_2x^m + \ldots$$

in the given equation, we obtain

$$x(1-x)[m(m-1)\,a_0x^{m-2}+(m+1)ma_1\,x^{m-1}+(m+2)\,(m+1)a_2x^m+\ldots]$$
$$-(1+3x)[ma_0x^{m-1}+(m+1)a_1x^m+(m+2)a_2x^{m+1}+\ldots]-[a_0x^m+a_1x^{m+1}+a_2x^{m+2}+\ldots]=0$$

Equating to zero the coefficients of the lowest power of x, we get $a_0[m(m-1)-m]=0$, $(a_0\neq 0)$,

$m(m-2)=0$, *i.e.* $m=0, 2$ *i.e.*, the two roots are distinct and differ by an integer.

Equating to zero the coefficients of successive powers of x, we get

$(m-1)a_1=(m+1)a_0$, $ma_2=(m+2)a_1$, $(m+1)a_3=(m+3)\,a_2$ and so on.

$$a_1=\frac{m+1}{m-1}\,a_0,\; a_2=\frac{(m+1)\,(m+2)}{(m-1)m}\,a_0, a_3=\frac{(m+1)\,(m+2)\,(m+3)}{(m-1)m(m+1)}\,a_0 \text{ etc.}$$

Thus (i) becomes

$$y=a_0x^m\left[1+\frac{m+1}{m-1}\,x+\frac{(m+1)\,(m+2)}{(m-1)m}\,x^2+\frac{(m+1)\,(m+2)\,(m+3)}{(m-1)m(m+1)}x^3+\ldots\right]$$

Putting $m=2$ (greater of the two roots) in (ii), the first solution is

$$y_1=a_0x^2\left[1+3x+\frac{3.4}{2}\,x^2+\frac{4.5}{2}x^3+\ldots\right]$$

If we put $m = 0$ in (*ii*), the coefficients become infinite.

To obviate this difficulty, put $a_0 = b_0 (m - 0)$ so that

$$y = b_0 x^m \left[m + \frac{m(m+1)}{m-1} x + \frac{(m+1)(m+2)}{m-1} x^2 + \frac{(m+1)(m+2)(m+3)}{(m-1)(m+1)} x^3 + \dots \right]$$

$$\frac{\partial y}{\partial m} = b_0 x^m \log x \left[m + \frac{m(m+1)x}{m-1} + \frac{(m+1)(m+2)}{m-1} x^2 + \frac{(m+1)(m+2)(m+3)}{(m-1)(m+1)} x^3 + \dots \right]$$

$$+ b_0 x^m \left[1 + \frac{m^2 - 2m - 1}{(m-1)^2} x + \frac{m^2 - m - 5}{(m-1)^2} x^2 + \frac{m^2 - 2m - 11}{(m-1)^2} x^3 + \dots \right]$$

$\therefore$ the second solution is $y_2 = \left(\frac{\partial y}{\partial m}\right)_{m=0}$

$$= b_0 \log x\, [-1.2x^2 - 2.3x^3 - 3.4x^4 - \ldots] + b_0[1 - x - 5x^2 - 11x^3 - \ldots]$$

Hence the complete solution is $y = c_1 y_1 + c_2 y_2$

$$y = \frac{1}{2} c_1 a_0 [1.2x^2 + 2.3x^3 + 3.4x^4 + \ldots] - b_0 c_2 \log x\, [1.2x^2 + 2.3x^3 + 3.4x^4 + \ldots]$$
$$- b_0 c_2 [-1 + x + 5x^2 + 11x^3 + \ldots]$$

$$y = (C_1 + C_2 \log x)(1.2x^2 + 2.3x^3 + 3.4x^4 + \ldots) + C_2(-1 + x + 5x^2 + 11x^3 + \ldots)$$

where $C_1 = \frac{1}{2} c_1 a_0,\ C_2 = -b_0 c_2$